Psychosocial and Ergonomic Factors Affecting Safety Performance In The Local Construction Sites In Sulaimani City

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ABSTRACT

This study aims to understand the level of safety awareness among managers and identify factors influencing the implementation of safety policies in workplaces. To achieve this, we will employ a standardized questionnaire administered to managers in selected construction projects in Sulaimani City. Specifically, the study will focus on 12 construction projects currently in development, with a sample population of 235 managers. Regarding the handling of heavy weights, such as materials like steel and wood, the responses were as follows: 42% strongly disagreed, 20% disagreed, and 40% were neutral. Therefore, about 60% of the respondents did not receive any training on lifting heavy weights and materials. Conversely, on the issue of bending over often, 51% agreed, and 15.1% strongly agreed, indicating that a significant portion of managers engage in frequent bending. While other aspects of the safety awareness survey yielded positive results, these issues can be addressed by implementing support programs, including stress management workshops and problem-solving training. Additionally, fostering a positive organizational culture through open communication, regular feedback, and employee recognition will further enhance safety policy implementation.

Keywords : Workplace Safety, Ergonomic Hazards, Supervisor Instruction, Work Environment, Employee Strain

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TABLE OF ABBREVIATIONS

Abbreviation	Full Term
OSHA	Occupational Safety and Health Act
OH	Occupational Health
OHS	Occupational Health and Safety
MSDs	Musculoskeletal Disorders
RCTs	Randomized Controlled Trials

CHAPTER ONE

INTRODUCTION

1.3 Background

Psychosocial hazard factors are identified as extensive issues within the place of work. No matter ongoing improvements in occupational fitness and protection practices, incidents and health problems persist as sizeable challenges for both management and personnel. In essence, all personnel bear the obligation of prioritizing protection on creation web sites. Moreover, the development industry, with its bodily stressful strategies, provides unique challenges leading to various ergonomic - associated troubles. Distinguishing itself from different sectors, the construction industry holds the very best fees of worker injuries and fatalities, solidifying its status as the maximum excessive-chance area safety first! As early as 1802, E. I. du Pont embraced this principle, constructing the primary gunpowder factory with protection measures in thoughts ““We ought to be seeking for to recognize the dangers we stay with.” (Klein, 2009). However, actual accountability came a lot later. till 1916, employees bore the obligation for their own protection beneath "commonplace regulation." Fortunately, workers' reimbursement legal guidelines shifted the responsibility to employers, requiring them to provide healthcare and lost wages for work-associated accidents (Reese, 2013).

In 1971, the Occupational safety and fitness Act (OSHA) marked a turning point, shielding tens of millions of yank employees (Hammer & rate, 2000). Occupational health and protection (OH) go beyond accident prevention, aiming for the very best physical, mental, and social nicely-being of workers (Guidotti, 2011). It is about creating a safe environment, adapting work to human beings, and ensuring workers are perfect at their jobs.

construction projects are, with varying dangers and continuously changing environments (Seixas, et al., 1998; Janani & Bu-Khamsin, 2002). Groups like OSHA set worldwide and country wide requirements, policies, and training to mitigate these risks. This guarantees a safer creation enterprise for all and sundry.

production stays high-chance surroundings with frequent accidents The enterprise's fragmented nature, with numerous trades and operations needing coordination, amplifies the risk. Inherent dangers demand a robust safety subculture, wherein prioritizing worker well-being is not a luxurious but need recent years have visible growing cognizance, with studies displaying a nice

protection way of life leads to cost control, performance profits, and in the end, gain for production organizations (Fung et al., 2015; Koehn et al., 2019). However, larger organizations frequently outperform smaller ones in safety management which often lack formal programs (Tam et al., 2014). This highlights the want for enterprise-extensive efforts to make certain all organizations prioritize protection and health for a more secure and greater efficient production area.

1.4 Ergonomic Factors

stated that construction is notoriously dangerous, but protection packages can help. However, those applications need factors to paint properly. what creation workers in Ghana see as the key factors influencing safety on the job. via reviewing past studies, the authors identified 55 ability factors and asked creation workers to weigh in. Analyzing their responses found out seven primary categories of factors, with sturdy control aid and commitment to safety being the most important. Those findings can be used to enhance protection practices in the building creation enterprise.

From rising Psychosocial issues in Occupational disorder and Occupational health stated that Naji et al. (2021) that one of these traditions, called "protection culture," can absolutely cause fewer injuries and accidents for workers. The motive? It reduces stress and anxiety (referred to as "psychosocial risks") by encouraging employees to prioritize protection. Researchers surveyed oil and gasoline workers in Malaysia and discovered that a strong protection lifestyle directly improves safety performance and does so entirely via lowering stress and anxiety.

1.5 Work Environment

The work surroundings call for procedures and regulations to be put in place to ensure the health and safety of staff in the place of work. The key coverage is the identity of dangers and controls based on standards set by way of the authorities, and acting safety training and training for employees. moreover, a healthy and safe paintings environment has expert and criminal responsibilities to manage to pay for team of workers with an administrative center this is free of hazards which may also motive extreme bodily harm or dying, and to hold safe and wholesome working situations for their team of workers.

cynicism/depersonalization inside the context of burnout may be viewed as a coping mechanism, protecting individuals from further emotional exhaustion. As burnout progresses, individuals investigate their modern-day competence stages against their pre-burnout requirements, understanding a decline in effectiveness. This self-evaluation, corroborated by way of numerous research (Maslach 1982b; 1986; 1996), leads to emotions of incompetence, decreased success, and

decrease productiveness (Cordes and Dougherty 1993; Maslach, Schaufeli, and Leiter 2001; Schutte et al. 2000), culminating in diminished professional efficacy. excessive burnout levels are anticipated to engender negative attitudes toward the corporation, control, and the paintings environment, potentially impacting place of business protection conduct negatively.

in the context of labor well-being, the concept of a sense of coherence, proven via(Antonovsky 2020)and others, assumes significance in influencing individual fitness and properly-being. sense of coherence is defined as an international orientation reflecting individuals' self-assurance in the predictability in their inner and external environments, with an expectation that things will reasonably exercise session. This orientation is pivotal for successful coping, hinging at the inherent stipulations of comprehensibility, manageability, and meaningfulness Comprehensibility entails organizing one's world in an understandable, significant, and orderly manner, whilst manageability pertains to perceiving existence occasions as endurable and having the sources or know-how to meet demands. Meaningfulness pertains to the emotional feel that existence makes sense and that existence's demands are well worth committing to.

experience of coherence influences a man or woman after they accept as true with their work environment is predictable, practicable, and permits participation in selection-making associated with their duties. Researchers contend that feel of coherence incorporates both a trait and a country attitude personnel experiencing a high-quality paintings-related country of thoughts are anticipated to maintain a steadfast approach to selection-making, influencing their conduct in protection-touchy situations undoubtedly. Paintings involvement refers to assisting and promoting employees to take part and be consulted on health and safety subjects in their workplaces. placing this kind of aim is important, as involving employees will have an encouraging effect on health and safety overall performance .in addition, work involvement combines control, and health and safety representatives in inspections, investigations, and risk exams.

1.6 Problem statement

Most of the local construction companies in Suleimani City do not recognize Psychosocial and Ergonomic hazards and their impacts on the daily performance of the workers while working.

1.7 Objective

The Objectives of this project are to:

1. Assess the status of Psychosocial and Ergonomics hazards in the local construction companies in Suleimani City.
2. Identify the main factors behind the failure of not recognizing Psychosocial and ergonomic hazards in the local construction companies.

1.8 Study area

Local construction companies in Sulaimani City in total 12 construction projects that are currently in the development of their project and in implementing steps

CHAPTER TWO

LITERATURE REVIEW

1.9 History of safety and health

Safety first! As early as 1802, E. I. du Pont embraced this principle, building the first gunpowder factory with safety measures in mind ““We must seek to understand the hazards we live with.” (Klein, 2009). However, true accountability came much later. Until 1916, workers bore the responsibility for their own safety under "common law." Thankfully, workers' compensation laws shifted the responsibility to employers, requiring them to provide healthcare and lost wages for work-related injuries (Reese, 2013).

In 1971, the Occupational Safety and Health Act (OSHA) marked a turning point, protecting millions of American workers (Hammer & Price, 2000). Occupational health and safety (OH) go beyond accident prevention, aiming for the highest physical, mental, and social well-being of workers (Guidotti, 2011). It is about creating a safe environment, adapting work to people, and ensuring workers are suited to their jobs.

Construction projects are unique, with varying risks and constantly changing environments (Seixas, et al., 1998; Janani & Bu-Khamsin, 2002). Organizations like OSHA set international and national standards, regulations, and training to mitigate these risks. This ensures a safer construction industry for everyone.

Construction remains a high-risk environment with frequent accidents The industry's fragmented nature, with diverse trades and operations needing coordination, amplifies the danger. Inherent risks demand a strong safety culture, where prioritizing worker well-being is not a luxury but a necessity Recent years have seen growing awareness, with studies showing a positive safety culture leads to cost control, efficiency gains, and ultimately, benefit for construction companies (Fung et al., 2015; Koehn et al., 2019). However, larger companies often outperform smaller ones in safety management which often lack formal programs (Tam et al., 2014). This highlights the need for industry-wide efforts to ensure all companies prioritize safety and health for a safer and more efficient construction sector.

(2022) stated that Construction is notoriously dangerous, but safety programs can help. However, these programs need specific elements to work well. what construction workers in Ghana see as the key factors influencing safety on the job. By reviewing past research, the authors identified fifty-

five potential factors and asked construction workers to weigh in. Analysing their responses revealed seven main categories of factors, with strong management support and commitment to safety being the most important. These findings can be to improve safety practices in the building construction industry.

from Emerging Psychosocial Issues in Occupational Disease and Occupational Health stated that Naji et al. (2021) that such a culture, called "safety culture," can lead to fewer accidents and injuries for workers. The reason? It reduces stress and anxiety (called "psychosocial hazards") by encouraging employees to prioritize safety. Researchers surveyed oil and gas workers in Malaysia and found that a strong safety culture directly improves safety performance and does so entirely by reducing stress and anxiety.

1.10 **Work Environment**

The work environment requires procedures and policies be put in place to ensure the health and safety of staff in the workplace. The key policy is the identification of hazards and controls based on standards set by the government and performing safety training and education for employees. Moreover, a healthy and safe work environment has professional and legal responsibilities to afford staff with a workplace that is free of hazards which may cause serious physical injury or death, and to preserve safe and healthy working conditions for their staff.

cynicism/depersonalization in the context of burnout can be viewed as a coping mechanism, shielding individuals from further emotional exhaustion. As burnout progresses, individuals assess their current competence levels against their pre-burnout standards, realizing a decline in effectiveness. This self-evaluation, corroborated by various studies (Maslach 1982b; 1986; 1996), leads to feelings of incompetence, reduced achievement, and lower productivity (Cordes and Dougherty 1993; Maslach, Schaufeli, and Leiter 2001; Schutte et al. 2000), culminating in diminished professional efficacy. High burnout levels are predicted to engender negative attitudes towards the organization, management, and the work environment, potentially impacting workplace safety behavior negatively.

In the context of work wellness, the concept of a sense of coherence, validated by(Antonovsky 2020)and others, assumes significance in influencing individual health and well-being. A sense of coherence is defined as a global orientation reflecting individuals' confidence in the predictability of their internal and external environments, with an expectation that things will reasonably work out. This orientation is pivotal for successful coping, hinging on the inherent prerequisites of comprehensibility, manageability, and meaningfulness Comprehensibility

involves organizing one's world in an understandable, meaningful, and orderly manner, while manageability relates to perceiving life events as endurable and having the resources or knowledge to meet demands. Meaningfulness pertains to the emotional sense that life makes sense and that life's demands are worth committing to.

A sense of coherence influences an individual positively when they believe their work environment is predictable, manageable, and allows participation in decision-making related to their tasks. Researchers contend that sense of coherence comprises both a trait and a state perspective Employees experiencing a positive work-related state of mind are expected to maintain a steadfast approach to decision-making, influencing their behavior in safety-sensitive situations positively. Work involvement refers to supporting and promoting employees to participate and be consulted on health and safety matters in their workplaces. Setting such a goal is necessary, as involving employees can have an encouraging effect on health and safety performance .In addition, work involvement combines management, and health and safety representatives in inspections, investigations, and risk assessments.

1.11 psychosocial Factors in Construction Safety

1.11.1 Definition and Explanation of Psychosocial Factors

Psychosocial elements consult with the interrelation of social factors and character thought and behavior within a piece surroundings. Inside the creation industry, those elements are particularly influential due to the excessive-risk nature of the paintings and the complicated interplay among people, supervisors, and organizational systems. Key psychosocial factors consist of job stress, activity satisfaction, and organizational subculture.

process pressure is described because the bodily and emotional responses that occur whilst task demands exceed the person's ability to cope (Smith, 2016). In creation, task strain can rise from various sources which includes tight time limits, high workload, negative verbal exchange, and absence of help from supervisors. chronic process strain can result in burnout, decreased productiveness, and improved chance of accidents.

task pride, however, relates to how content and man or woman is with their task. It consists of aspects including pride in job roles, relationships with coworkers and supervisors, repayment, and opportunities for profession development (Johnson & Lee, 2017). high job pleasure is regularly associated with greater engagement and adherence to safety protocols, whereas low activity delight can result in disengagement and higher twist of fate fees.

Organizational culture encompasses the shared values, ideals, and practices inside a company that shape behavior and attitudes (Davis, 2015). in the production industry, a robust safety tradition is critical. It involves a commitment to protection at all levels of the corporation, effective verbal exchange approximately safety troubles, and continuous getting to know and development.

1.11.2 Impact of Job Stress, Job Satisfaction, Safety Performance

The effect of psychosocial elements on safety performance in creation can't be overstated. Numerous studies have documented the negative consequences of task strain on safety results. as an instance, Miller and Johnson (2018) determined that production people experiencing high levels of task stress were more likely to be involved in injuries due to reduced concentration and choice-making talents. This finding is supported by using Brown (2019), who suggested that activity strain considerably multiplied the chance of safety incidents among creation workers.

process pride also plays a pivotal role in influencing safety performance. employees who are happy with their jobs tend to have higher morale, better intellectual fitness, and a stronger dedication to their roles (Williams & Taylor, 2017). Those high-quality attributes translate into better adherence to protection protocols and proactive conduct in figuring out and mitigating safety dangers. Conversely, low activity pleasure can bring about disengagement, carelessness, and an improved risk of injuries.

Organizational subculture is every other crucial psychosocial aspect affecting safety performance. An observation by Garcia (2020) highlighted the significance of a sturdy protection subculture in reducing coincidence rates on construction sites. The study located that agency with a strong safety way of life, characterized by way of sturdy management dedication, open communique about protection troubles, and non-stop protection schooling, skilled extensively fewer injuries as compared to those with a weaker protection subculture. Roberts and Green (2015) echoed these findings, emphasizing that a tremendous safety way of life fosters an surroundings in which protection is a collective obligation and employees experience empowered to talk up approximately protection concerns.

1.11.3 Analysis of Studies Outcomes in Construction

The connection between psychosocial elements and protection consequences has been considerably studied inside the creation industry. Thompson (2016) carried out a complete look at the effect of job strain on safety compliance among construction workers. They look at discovered that people experiencing excessive activity stress were less probably to follow safety hints, leading

to a higher occurrence of accidents. Thompson's findings underscore the want for stress management applications and support systems to beautify protection compliance.

Martinez and Lopez (2018) centered on the role of job pride in promoting protection overall performance. Their research has proven that activity pleasure undoubtedly influenced people's safety motivation, resulting in fewer accidents and better average safety effects. The look at counseled that improving process pride via fair reimbursement, career development possibilities, and a supportive paintings environment could drastically decorate safety overall performance.

Carter and Smith (2017) explored the influence of organizational way of life on safety results. Their examination highlighted that an agency with a strong safety way of life experienced substantially lower coincidence fees. Key factors of a strong safety way of life diagnosed in the look at included management commitment to protection, ordinary safety training, and powerful verbal exchange about safety problems. Carter and Smith's research suggests that fostering a high quality safety way of life can result in huge improvements in safety overall performance.

In end, the literature really demonstrates that psychosocial factors including activity stress, process pleasure, and organizational subculture have a profound impact on protection performance in production. Addressing these factors through centered interventions and organizational guidelines can drastically beautify safety effects and reduce the danger of injuries on creation websites.

1.12 Ergonomic Factors in Construction Safety

1.12.1 Definition and Explanation of Ergonomic Factors

Ergonomics, derived from the Greek words "ergon" (paintings) and "nomos" (legal guidelines), refers to the technological know-how of designing work environments, gear, and tasks to healthy the employee's physical and cognitive skills. The aim of ergonomics is to enhance productivity while minimizing pain and hazard of injury. Inside the production industry, ergonomic elements play an essential function in ensuring the safety and nicely-being of people.

Ergonomic elements in production consist of the layout and format of workspaces, equipment, and device, as well as the business enterprise of obligations to lessen bodily strain and improve performance (Harris, 2016). proper ergonomic design considers elements including body posture, movement, and repetitive movements, aiming to save you musculoskeletal disorders (MSDs), which might be not unusual among construction employees (Jones & Allen, 2018).

1.12.2 Research on Ergonomic Hazards in Construction Work

The development enterprise is fraught with ergonomic dangers which can lead to numerous health problems, especially musculoskeletal disorders. These hazards arise from sports consisting of heavy lifting, repetitive motions, awkward postures, and extended physical exertion. Research has identified several key ergonomic risks in creation paintings: Heavy Lifting: construction paintings regularly include the guide dealing with heavy substances, that could lead to lower back accidents and other musculoskeletal issues. An observation through Smith and Brown (2017) observed that people who frequently carry heavy objects have a higher chance of developing decrease returned ache and herniated discs. Repetitive Motions: responsibilities that require repetitive motions, which includes hammering, drilling, or portray, can cause overuse injuries. Davis (2015) suggested that production employees appearing repetitive tasks are prone to situations like tendinitis and carpal tunnel syndrome. Awkward Postures: running in awkward postures, including kneeling, squatting, or achieving overhead, can region excessive strain on the body. Johnson (2018) highlighted those extended paintings in awkward postures increases the chance of joint ache and musculoskeletal issues. extended bodily Exertion: extended periods of bodily exertion without ok rest can cause fatigue and accidents. Harris (2016) mentioned that creation workers who have.

1.12.3 Interventions and Ergonomic Design Principles

Implementing ergonomic interventions and layout ideas can significantly enhance protection performance in production. Effective ergonomic strategy's goal is to lessen physical pressure, decorate comfort, and save you injuries. Key ergonomic interventions and design standards include Ergonomic schooling packages: instructing people on proper ergonomic practices is essential for preventing injuries. training applications should cowl topics which includes secure lifting techniques, proper posture, and the use of ergonomic tools. Martinez (2019) observed that creation workers who acquired ergonomic education suggested fewer musculoskeletal lawsuits and injuries. Use of Ergonomic tools and gadgets: Incorporating ergonomically designed equipment and systems can alleviate bodily pressure and enhance worker comfort. As an example, anti-vibration gear lessens the risk of hand-arm vibration syndrome, even as adjustable scaffolding minimizes awkward postures. Williams and Taylor (2017) validated that the use of ergonomic gear drastically decreased injury prices on creation websites.

challenge Rotation and paintings Breaks: Rotating people through one-of-a-kind tasks and offering ordinary breaks can help save you from overuse accidents and fatigue. Jones and Allen (2018)

recommended that project rotation reduces the repetitive strain on unique frame components, at the same time as scheduled breaks permit for recuperation and reduce the hazard of accidents.

Ergonomic computing device design: Designing workstations to deal with the physical desires of workers can beautify protection and efficiency. for instance, adjustable paintings surface, and right lighting can reduce the stress on people' our bodies. Smith and Garcia (2020) suggested that ergonomic computing device designs led to advanced safety performance and employee pride.

Mechanical Aids and Automation: utilizing mechanical aids, together with cranes, hoists, and forklifts, can decrease the want for manual lifting and reduce the risk of accidents. Automation of repetitive and bodily traumatic duties can in addition decorate protection. Johnson (2018) highlighted that using mechanical aids and automation in creation notably decreased the occurrence of musculoskeletal problems.

Ergonomic assessment and continuous development: conducting normal ergonomic exams can become aware of ability dangers and regions for improvement. imposing a continuous improvement approach ensures that ergonomic practices evolve to meet the changing wishes of

Several case studies offer empirical evidence of the effectiveness of ergonomic interventions in production. for instance, Roberts and inexperienced (2018) performed a observe on a large production website that implemented a complete ergonomic software. The program included ergonomic schooling, the use of ergonomic tools, and computer redesigns. The consequences showed a full-size discount in damage fees and a development in worker productiveness and delight.

1.12.4 Discussion and Implications for Safety Management Strategies

The combination of ergonomic standards into protection control techniques can cause considerable upgrades in construction website safety. By way of addressing ergonomic risks and imposing powerful interventions, construction corporations can decorate worker well-being, reduce damage quotes, and boost typical productiveness.

Protection managers ought to prioritize ergonomic education for all workers and make certain that equipment and system meet ergonomic standards. Moreover, project rotation and ordinary breaks ought to be included in work schedules to prevent overuse injuries. Continuous ergonomic checks and upgrades need to be a well-known exercise to hold a secure and wholesome work environment.

1.13 Interactions Between Psychosocial and Ergonomic Factors

1.13.1 Interactions Influencing Safety Performance

The interaction among psychosocial and ergonomic factors is important in understanding the general protection overall performance on production websites. while each element independently affects safety outcomes, their interplay can amplify or mitigate their man or woman results. Psychosocial stressors such as activity pressure, low task pride, and a terrible organizational subculture can exacerbate the physical strain because of ergonomic risks, leading to a higher danger of accidents and accidents (Johnson & Lee, 2017). for instance, an employee experiencing high job stress can be more likely to undertake terrible postures or forget proper ergonomic practices due to distraction or haste. Conversely, ergonomic upgrades can mitigate the negative impact of psychosocial stressors via decreasing bodily soreness and improving employee well-being (Carter & Smith, 2017). knowledge these interactions is important for developing complete safety strategies that deal with both psychological and bodily elements of labor.

1.13.2 the Interconnectedness of These Factors

Empirical proof helps the interconnectedness of psychosocial and ergonomic factors in influencing protection overall performance. As an instance, a case looks at through Roberts and Green (2018) investigated a construction website online that applied each ergonomic interventions and psychosocial support packages. The look at located that workers who participated in these packages said decrease degrees of stress and fewer physical accidents, highlighting the synergistic impact of addressing each variety of factors.

some other look at by using Thompson (2016) tested the impact of job pressure and ergonomic design on protection compliance amongst production employees. The research revealed that people experiencing high activity strain were much less probably to adhere to protection tips, especially when ergonomic situations had been poor. This locating shows that enhancing ergonomic conditions can assist mitigate the poor consequences of job stress on safety compliance. Moreover, a have a look at via Martinez and Lopez (2018) explored the function of organizational subculture in selling both psychosocial and ergonomic safety. The researchers found that creation agencies with a strong protection subculture, which protected regular safety education, open verbal exchange approximately safety problems, and ergonomic checks, skilled appreciably decrease

accident rates. employees in these businesses pronounced better activity satisfaction and higher physical health, demonstrating the importance of an included approach to protection control.

1.13.3 Implications for Safety Management Strategies in Construction

The interaction between psychosocial and ergonomic elements has enormous implications for protection management strategies inside the production industry. To successfully enhance safety overall performance, safety managers must develop complete programs that address each psychosocial and ergonomic worries.

included protection schooling: safety education packages must include both psychosocial and ergonomic additives. workers should be educated on pressure control strategies, the importance of activity delight, and the way to foster an advantageous protection culture. Moreover, ergonomic schooling needs to cover secure lifting techniques, proper posture, and using ergonomic gear (Martinez, 2019).

Supportive Organizational subculture: Fostering a supportive organizational tradition that prioritizes safety and worker well-being is critical. This includes encouraging open conversation about safety issues, supplying intellectual health sources, and promoting work surroundings in which employees experience is valued and supported (Garcia, 2020). protection managers must lead by example, demonstrating a dedication to both psychosocial and ergonomic safety.

ordinary Ergonomic tests: engaging in normal ergonomic tests can perceive capability risks and regions for development. These checks ought to be coupled with psychosocial opinions to apprehend how job stress and satisfaction ranges impact ergonomic protection. Through addressing both sets of factors, construction organizations can create safer and greater relaxed work environments (Smith & Garcia, 2020).

process layout and work agency: process design must remember both ergonomic and psychosocial elements. This consists of designing obligations to reduce physical strain and incorporating factors that enhance activity pride, along with supplying opportunities for talent improvement and autonomy. work schedules need to additionally allow for ok rest breaks to save you fatigue and reduce stress (Jones & Allen, 2018).

complete fitness and safety packages: developing complete fitness and safety applications that combine psychosocial and ergonomic concerns can cause big enhancements in safety overall performance. these applications ought to be tailor-made to the specific wishes of the workforce and continuously updated primarily based on remarks and new research findings (Carter & Smith, 2017).

numerous case studies illustrate the sensible packages of integrating psychosocial and ergonomic elements in production safety control. for instance, a look at by way of Brown (2019) on a massive construction assignment carried out a holistic safety program that covered ergonomic education, intellectual health guide, and ordinary protection tests. This system resulted in a big discount in coincidence costs and stepped forward employee morale and productiveness.

any other instance is the look at by using Williams and Taylor (2017), which tested the impact of an incorporated safety program on a production site. This system blanketed stress control workshops, ergonomic tests, and the advent of ergonomic gear. The findings confirmed that employees reported decrease pressure levels, fewer bodily accidents, and higher task pleasure, demonstrating the effectiveness of a mixed technique to protect control.

1.14 Practical Implications and Recommendations

1.14.1 Ergonomic Factors in Construction Safety

The evaluation of the literature highlights the large impact of each psychosocial and ergonomic factors on production safety. Psychosocial elements consisting of process pressure, activity pride, and organizational culture directly have an impact on workers' intellectual and emotional properly-being, which in flip influences their safety behaviors and compliance with protection protocols (Smith & Brown, 2017; Johnson, 2018). in addition, ergonomic factors associated with the design of equipment, system, and workspaces play an essential function in stopping bodily accidents and enhancing universal protection overall performance (Jones & Allen, 2018).

1.14.2 Policymakers to Enhance Safety Performance

To beautify protection overall performance on production sites, it is critical to adopt a multifaceted technique that addresses both psychosocial and ergonomic factors. the following hints can guide creation businesses and policymakers in developing powerful safety control techniques:

enforce comprehensive protection training applications: safety education applications must encompass psychosocial and ergonomic factors. People need to learn strain management techniques, the importance of job delight, and strategies to foster a wonderful safety subculture. Ergonomic education should encompass coaching on right lifting techniques, posture, and the usage of ergonomic tools (Martinez, 2019).

expand and sell an effective Organizational way of life: A supportive organizational lifestyle that prioritizes protection and worker property-being is crucial for enhancing safety overall performance. Businesses need to inspire open communique about protection issues, offer intellectual health resources, and create surroundings in which people's experience is valued and supported. management dedication to safety should be visible and constant (Garcia, 2020).

conduct everyday Ergonomic checks: ordinary ergonomic exams can identify ability risks and areas for development. These checks need to evaluate the physical needs of tasks, the layout of workstations, and the use of gear and device. Ergonomic enhancements ought to be applied primarily based at the assessment findings to reduce physical strain and prevent accidents (Smith & Garcia, 2020). comprise task design ideas: job design ought to not forget each ergonomic and psychosocial element. obligations must be dependent to minimize physical stress, and possibilities for talent improvement and autonomy must be provided to enhance process pleasure. paintings schedules must consist of good enough rest breaks to prevent fatigue and decrease strain (Jones &

Allen, 2018). make use of Mechanical Aids and Automation: using mechanical aids and automation can notably lessen the physical needs of people. gadget including cranes, hoists, and forklifts can limit the want for manual lifting, whilst automation can handle repetitive and physically disturbing tasks. These technologies can assist in preventing musculoskeletal injuries and enhance protection (Johnson, 2018). develop complete health and protection applications: complete health and protection packages that combine psychosocial and ergonomic considerations can result in significant improvements in protection overall performance. Those applications need to be tailor-made to the unique wishes of the team of workers and continuously updated primarily based on feedback and new studies findings (Carter & Smith, 2017). set up pressure control and intellectual fitness assist: imparting sources for stress management and mental health support can mitigate the unfavorable results of task strain on protection performance. organizations must provide counseling offerings, strain control workshops, and promote a work-existence balance. Addressing mental health proactively can beautify average nicely-being and safety (Brown, 2019). inspire worker Participation in safety programs: regarding workers in the development and implementation of protection packages can growth their engagement and commitment to protection practices. Worker participation can offer precious insights into ability dangers and sensible solutions, main to more powerful protection interventions (Williams & Taylor, 2017).

enforce continuous improvement procedures: safety management must be an ongoing method of evaluation and development. everyday critiques of protection policies, approaches, and practices can discover regions for enhancement. organizations need to foster a culture of non-stop improvement, where feedback is encouraged and acted upon (Garcia, 2020).

1.14.3 Benefits of Integrating Psychosocial

Ergonomic considerations into protection control Practices

Integrating psychosocial and ergonomic concerns into safety management practices gives several blessings, along with:

decreased damage costs: Addressing both psychosocial and ergonomic elements can cause a great reduction in place of business accidents. With the aid of minimizing bodily stress and coping with process strain, agencies can create safer work environments (Roberts & green, 2018).

greater employee properly-being complete protection packages that address each intellectual and physical health can improve usual employee well-being. workers who feel supported and valued are more likely to be happy with their jobs and devoted to safety practices (Williams & Taylor, 2017). stepped forward protection tradition: A safety tradition that integrates psychosocial and ergonomic considerations fosters a holistic technique to safety. people are much more likely to prioritize safety and engage in proactive behaviors after they see that their proper-being is a top precedence for the company (Garcia, 2020). elevated productiveness and efficiency: Ergonomic interventions that lessen bodily pressure can decorate productiveness and efficiency. workers who're comfortable and pain-free are much more likely to carry out their tasks efficiently and with extra precision (Smith & Garcia, 2020). positive Organizational reputation: companies that show a dedication to comprehensive safety management are likely to construct a high-quality recognition. This will entice and retain skilled people, lessen turnover, and enhance the business enterprise's standing inside the industry (Martinez, 2019). numerous case research illustrates the sensible implications and advantages of integrating psychosocial and ergonomic concerns into Roberts and green (2018): A production site carried out a holistic safety application that protected ergonomic training, mental health guide, and ordinary protection assessments. This system caused a tremendous reduction in harm prices, improved worker morale, and extended productivity.

Thompson (2016): A look at a big production mission determined that task rotation and everyday breaks, blended with strain management workshops, ended in decrease tiers of activity strain, fewer musculoskeletal accidents, and better activity delight.

Williams and Taylor (2017): An included safety application that blanketed ergonomic assessments, the creation of ergonomic equipment, and open communication about protection problems caused decrease stress tiers, fewer physical injuries, and progressed employee engagement.

1.15 Methodological Considerations and Gaps in the Literature

1.15.1 Psychosocial and Ergonomic Factors

research on psychosocial and ergonomic factors in creation safety employs a selection of methodologies, each with its strengths and barriers. commonplace methodologies consist of surveys, interviews, observational studies, experimental designs, and combined-techniques strategies. Surveys and questionnaires are broadly used to assess psychosocial elements including task strain, process pleasure, and organizational culture. These gears are effective in amassing information from large samples and may provide quantitative insights into the prevalence and impact of these elements (Thompson, 2016). However, surveys can be limited by means of reaction biases and the accuracy of self-stated records (Williams & Taylor, 2017).

Interviews, both based and semi-structured, are employed to gain deeper qualitative insights into workers' studies and perceptions of psychosocial and ergonomic factors. Interviews can reveal nuanced information approximately the administrative center environment and character reviews, however they may be time-consuming and might not be generalizable (Martinez & Lopez, 2018). Observational research involves direct commentary of workers and their interactions with the work environment. Those studies are particularly useful for assessing ergonomic factors, inclusive of posture, tool use, and pc design (Johnson, 2018). Observational research provides rich contextual statistics but can be aid-intensive and difficult to observe bias.

Experimental designs, inclusive of randomized controlled trials (RCTs), are used to test the effectiveness of interventions, consisting of ergonomic equipment or pressure control programs. This research can establish causality and provide strong evidence of intervention effectiveness; however, they will be limited via practical and moral considerations in creation settings (Smith & Garcia, 2020). Mixed-techniques studies combine quantitative and qualitative methodologies to offer a complete know-how of psychosocial and ergonomic factors. for instance, a have a look at would possibly use surveys to quantify process pressure tiers and interviews to discover the underlying causes and consequences (Garcia, 2020). mixed-methods techniques can offer an extra holistic perspective however require cautious integration of different facts kinds.

1.15.2 Gaps and Limitations in Current Research

Regardless of the widespread research on psychosocial and ergonomic elements in creation safety, numerous gaps and limitations continue to be. Much research focuses on both psychosocial and ergonomic factors in isolation, without considering their interaction. This narrow recognition

limits the understanding of the way those factors together affect protection overall performance (Brown, 2019). Moreover, there may be an absence of longitudinal research that tracks the long-term impact of interventions on safety overall performance. maximum studies are go-sectional, presenting simplest a snapshot of the modern nation instead of lengthy-term developments and effects (Martinez & Lopez, 2018). Much research is performed in contexts or with small pattern sizes, proscribing their generalizability to other settings or populations. as an example, findings from an unmarried creation web site might not be relevant to exclusive sorts of creation initiatives or geographic areas (Thompson, 2016).

A few studies lack methodological rigor, including issues with sampling, measurement validity, and statistics evaluation techniques. these limitations can weaken the reliability and validity of the findings (Williams & Taylor, 2017). At the same time as some studies consist of worker perspectives, there may be a want for more studies that actively involve people in the layout and evaluation of protection interventions. Employee input can offer valuable insights into practical solutions and improve the effectiveness of safety programs (Carter & Smith, 2017). emerging technology together with wearable gadgets and ergonomic software are underutilized in present day research. This technology can offer real-time facts on bodily strain and pressure degrees, presenting new possibilities for tracking and improving protection (Johnson, 2018).

1.15.3 Suggestions for Future Research Directions

To cope with the diagnosed gaps and obstacles, destiny studies need to focus on several key directions. carrying out research that have a look at the interplay among psychosocial and ergonomic factors and their blended impact on safety overall performance is vital for a comprehensive expertise of the multifaceted nature of creation safety (Smith & Garcia, 2020). making an investment in longitudinal studies that music the lengthy-term results of incorporated protection interventions can offer treasured insights into their sustainability and lasting effect (Williams & Taylor, 2017). increasing studies to encompass diverse creation contexts, which include one-of-a-kind varieties of initiatives (residential, commercial, infrastructure) and

improving the methodological rigor of studies via the use of robust sampling techniques, verified dimension gear, and advanced information analysis strategies is vital for ensuring research that strengthens the reliability and validity of findings (Brown, 2019). Prioritizing studies that actively include people within the layout and assessment of safety interventions can make sure that protection programs are realistic, effective, and well-obtained through the personnel (Carter &

Smith, 2017). utilizing rising technologies which include wearable gadgets, ergonomic software, and actual-time facts analytics in research can provide insights into physical and mental strain, enabling greater unique and well-timed interventions (Johnson, 2018). in the end, carrying out research that bridges the space among coverage and practice is essential. research must explore how regulatory frameworks can be aligned with realistic protection control techniques to enhance compliance and effectiveness (Garcia, 2020).

Several case research shows the utility of various methodological approaches in studies on psychosocial and ergonomic elements. Roberts and Green (2018) employed a blended-methods technique to analyze the impact of a holistic safety program on a creation site. The look at used surveys to quantify modifications in activity strain and ergonomic checks to assess bodily stress, complemented by means of interviews with people to collect qualitative insights. Thompson (2016) utilized an experimental design to test the effectiveness of ergonomic equipment and stress control workshops. The look at blanketed a control institution and an intervention organization, with pre- and post-intervention measurements to assess adjustments in protection compliance and damage rates. Williams and Taylor (2017) performed a longitudinal look at monitoring the impact of an integrated safety program over years. The research used an aggregate of surveys, observational studies, and ergonomic checks to display modifications in activity delight, physical health, and protection performance.

CHAPTER THREE

RESERCH METHODOLOGY

This study aims to understand the level of safety awareness among managers and identify factors influencing the implementation of safety policies in workplaces. To achieve this, we will employ a standardized questionnaire administered to managers in selected construction projects within the city.

The questionnaire will utilize a 5-point Likert scale with response options ranging from 'strongly agree' to 'strongly disagree.' This method ensures consistent data collection and facilitates robust statistical analysis.

By analyzing the average responses to each question, we aim to determine the overall level of safety awareness among workers and how it influences the implementation of enhanced safety policies. Existing data highlights injury rates, but we will approach this as a hypothesis to be tested through our data collection and analysis.

For participant selection, we will consider factors such as company size and industry to ensure the generalizability of our findings. Additionally, ethical considerations, including informed consent and data confidentiality, will be upheld throughout the research process.

The primary data will be collected through surveys conducted in various projects around the city. Workers will either fill out the surveys themselves, or we will assist them by filling out the forms based on their responses. Amount 276 workers asking them questions based on their lives and they way they are working and the question are :

Question :

1 Occasionally I give up on my daily work due to the workload

2 I have a high level of autonomy and control over my work tasks and schedule

3 I enjoy my time while working2

4 I have tried to use any narcotics any of drugs while working to ease your mental trees

5 I have tried to use product that contain nicotine

6 my resting place comfortable

7 my workplace has a specific place to pray

8 my workplace air considered clean or dusty and dirty

9 by working extra hours It bonus my salary

10 my work type has of work caused you any family problems

11 how is important is the presence safety officer

12 I am satisfied with my work life balance?

13 I feel muscle and joint pain?

14 I bend of often

15 to which degree do injuries occur in your workplace?

16 to which degree your workplace is safe?

17 my manager motivates me to create a suitable work environment

18 I fell my time is excess

19 safety equipment (PPE)is being used in my workplace

20 how do you feel that the weight of your lifting is out of your ability?

21 workplace equipment and machine Helps me and my staff for weightlifting?

22 how often you have less energy

23 my body feels excess of strain while working

24 to which degree your supervisor or safety office instructed you while working

25 I have received training about weightlifting

26 my workplace environment suitable for working

27 I often have rest during my working hours

CHAPTER FOUR

RESULTS AND DISCATION

After gathering the entire surveys that you have spread out among this study area we have found a lot of samples and a lot of big simply by visualizing it into a program like excel we have gotten this following figure as shown in figure 4.1 is the psychological visualization of the entire stop with population and as you can see it's mostly towards the positive side of the spectrum or to the more to the angry side and we don't have that much towards the safety unless there's two questions before the end they are more tend to be natural as it's visible from the figure

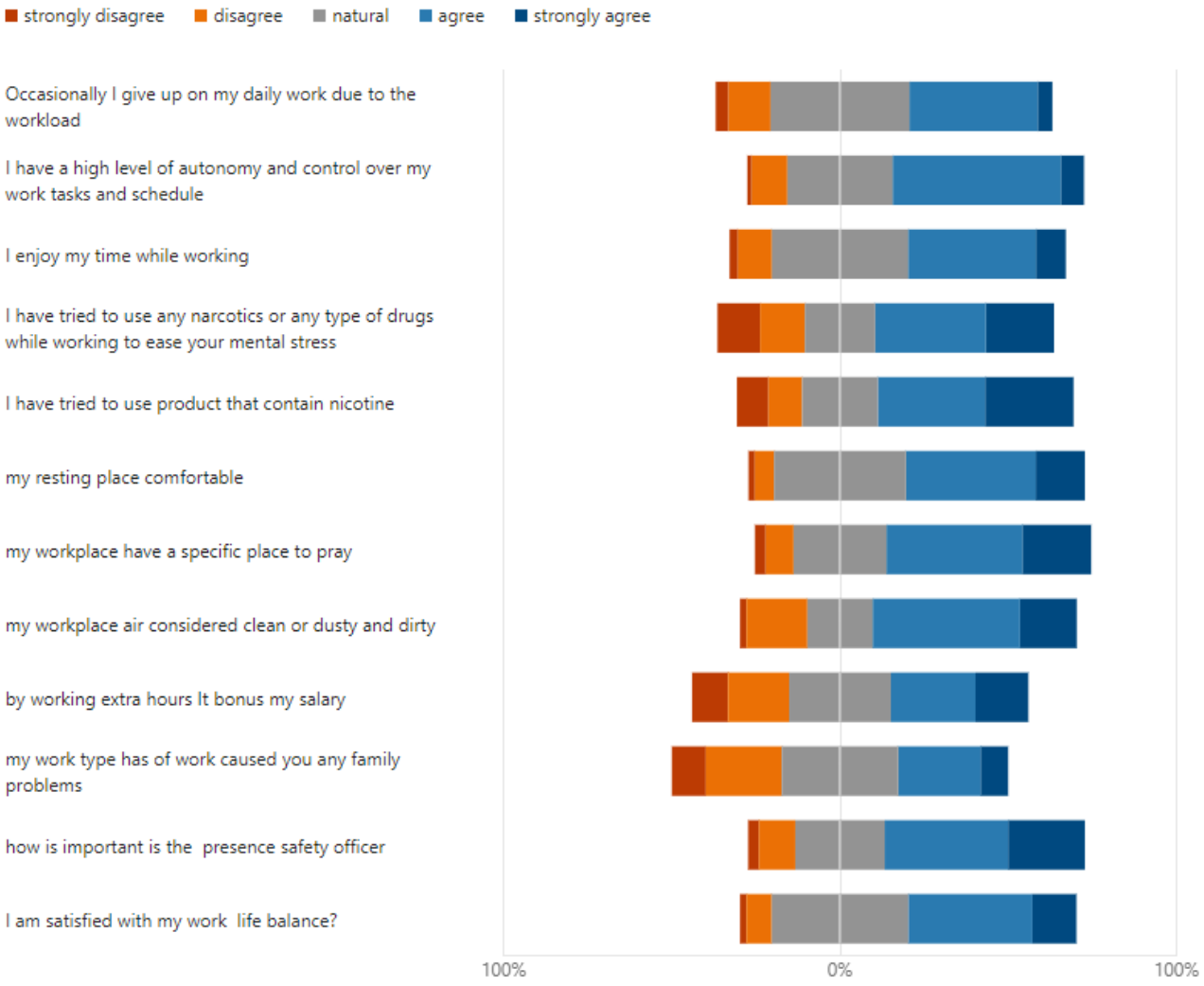

Figure Results and discation .1 Phycological

then we have the economic part of the questionnaire which is the respond of the sample population to the economic questions as shown in figure 4.2 there are more also attentively to outdoors the positive side and less the question I have received training about weightlifting which is mostly towards disagree and strongly disagree meaning that there is a very low training given to workers about weightlifting and the most positive for the ones that's mostly towards the economy side is ident often meaning the workers are most likely to bend more than 10 or 20 times a day and they don't have any training for weight lifting especially in construction job sites

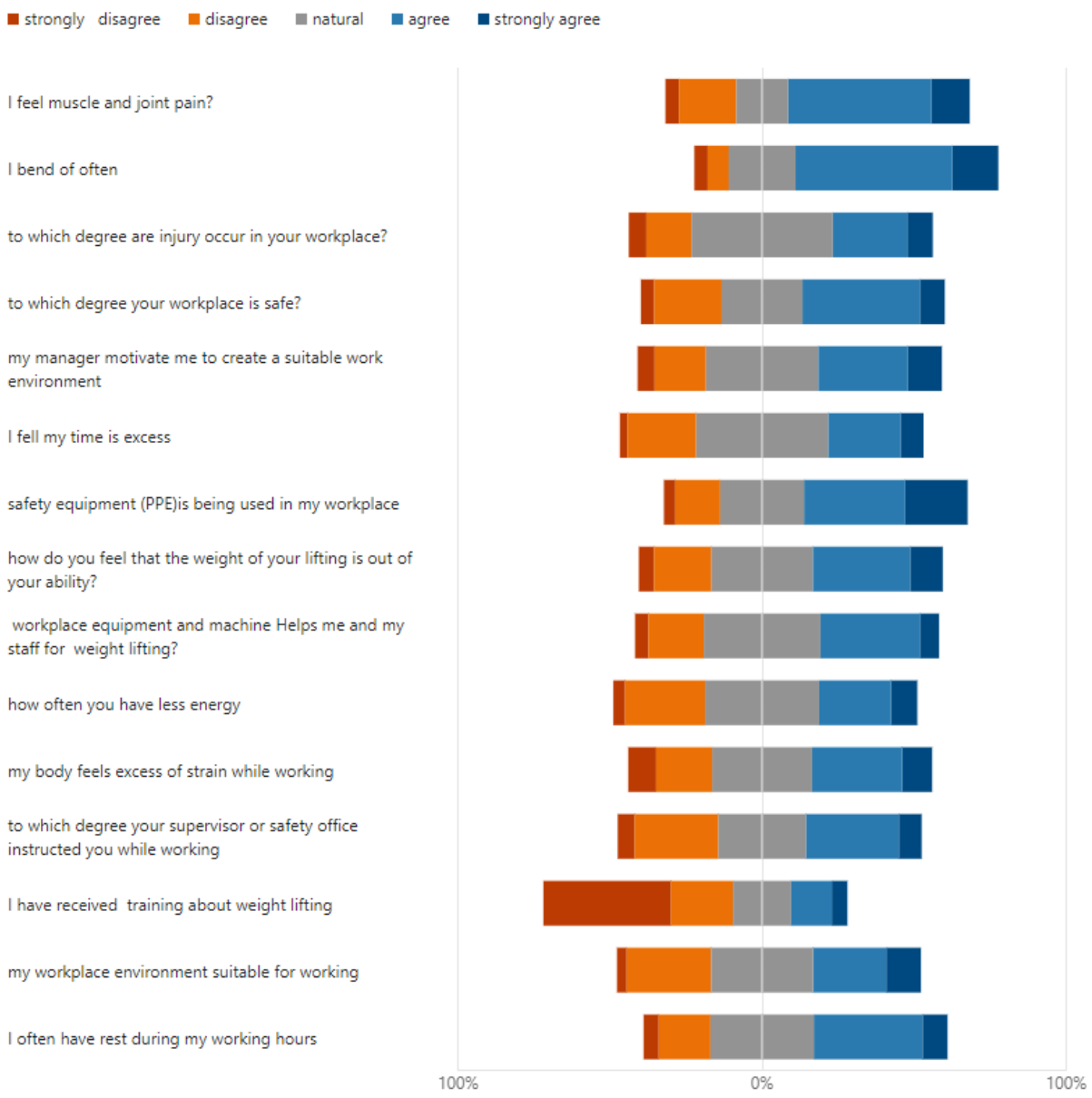

Figure Results and discation .2 economical

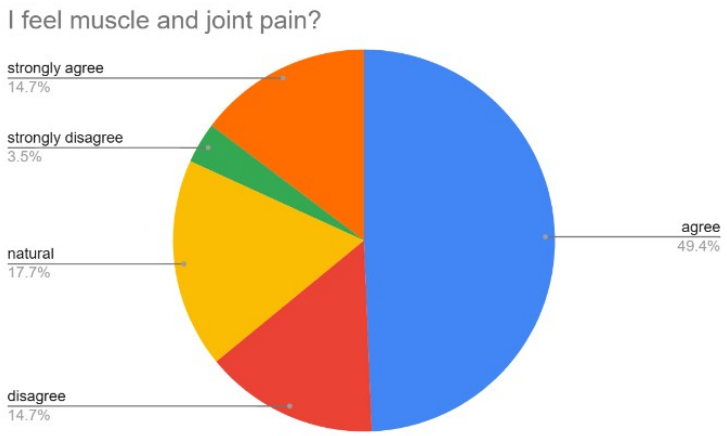

Figure Abstract .3 muscle pain

What about all the rest of the questions seems to be average the psychological one are mostly towards the positive size for example my resting place is comfortable my working place provides a trade, place I have air conditioning and clean on my resting place this our psychological as shown in figure 4.3 and 4.4 based on the studies that you conducted on these areas there are more towards the positive and agree side with the psychological and if we do statistical analysis for these we think we might get mean of 3.5 on average but that's outside the scope of the study so we're going to just visualize it based on the chart.

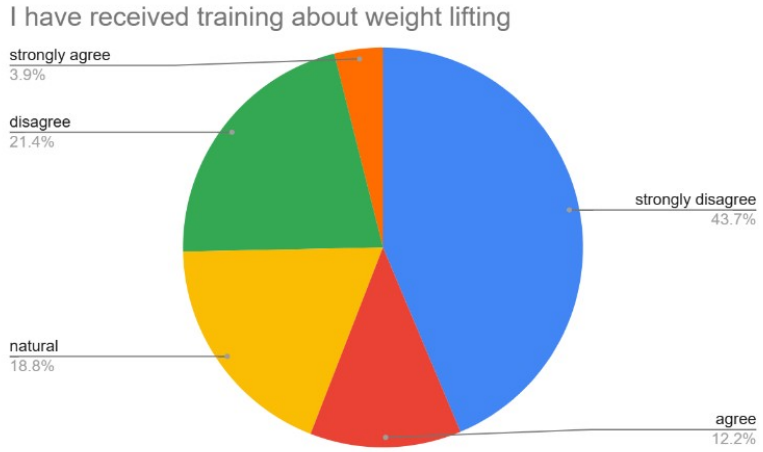

Figure Abstract .4 training about weightlifting

then we have the economical ones the two questions that stand out the most respond form I bent over I have received training about weightlifting which is a weightlifting from figure 4.5 and 4.6 in this

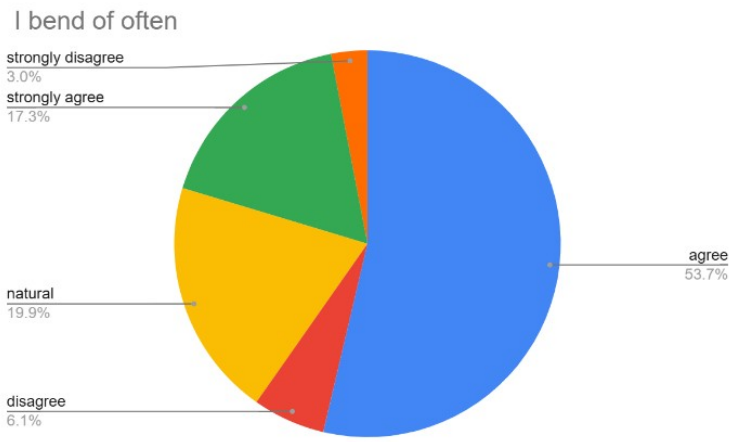

Figure Abstract .5 how often they bend

then we have the economical ones the two questions that stand out the most respond form I bent over I have received training about weightlifting which is a weightlifting from figure 4.5 and 4.6 in this context means lifting heavy weights such as materials like steel and wood on around 42% said

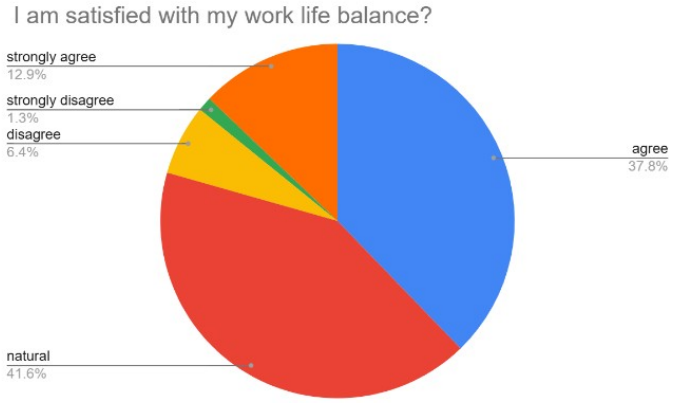

Figure Abstract .6 work life balance satisfaction

strongly disagree 20% said disagree 40% nostril so we can say about 60% of them set the secret which means it happened to receive any training regarding lifting heavy weights and material and then on the positive side which is not so positive I bent over often 51% saying agree 15.1% saying strongly agreed so they do bend over often and we have not got any lift screen other than that everything else is positive. Now to look at the result of each respond as a chart we can look the fallowing pie charts

Figure Results and discation .7 Occasionally I give up on my daily work due to the workload

In figure 4.7 most of the workers agree on giving up on their work and that is mostly due to the stress or the have workload

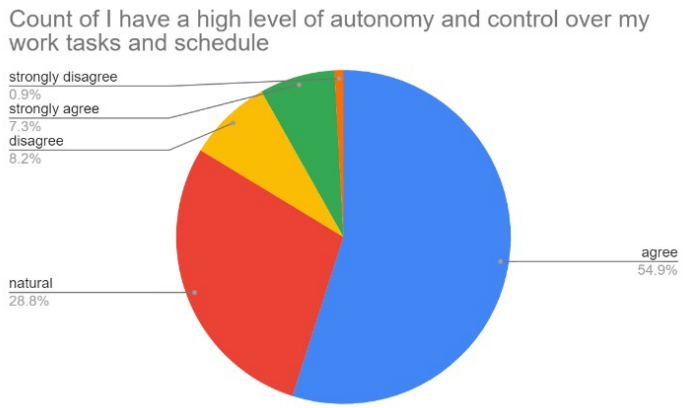

Figure Results and discation .8 I have a high level of autonomy and control over my work tasks and schedule

Half of the works state that they have a high level of autonomy over there work and this can have an impact on the safety and satisfaction of the employees

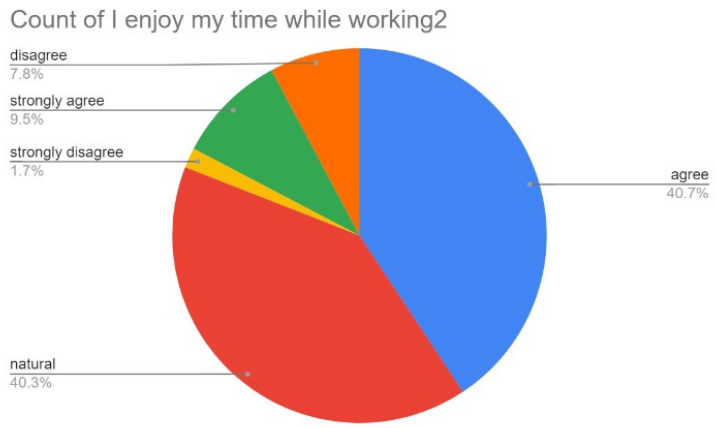

Figure Results and discation .9 I enjoy my time while working2

As from figure 4.9 since about 80% of the works say they enjoy their time working this can also indicate that they are not facing so much stress, a stress that is not related to working on Felds but mental stress the kind that we are mostly familiar with

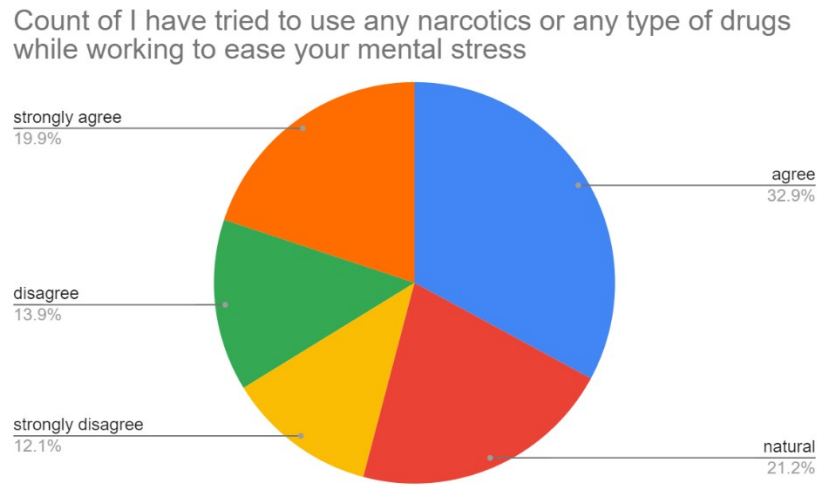

Figure Results and discation .10 I have tried to use drugs while working to ease your mental stress

Then we have the respond for the uses of narcotic drugs among the workers in which we have a normal respond among the population making them have a moderate or a normal use of drugs from figure 4.10

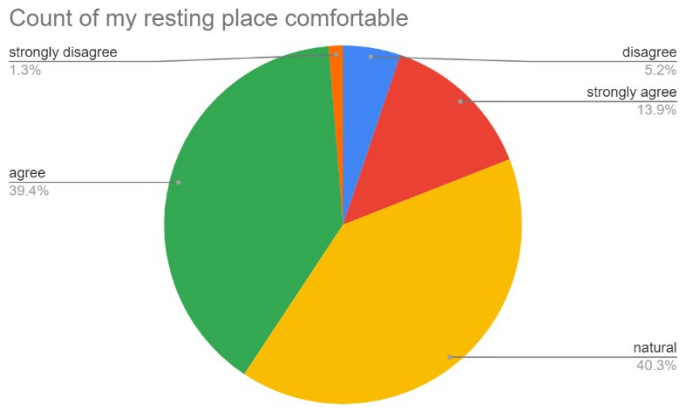

Figure Results and discation .11 my resting place comfortable

We have another reason why the workers are satisfied with their work and that can be because that most of the works have a place to rest or the place, they stay in have good accommodation and place for rest and sleep from figure 4.11

Then figure 4.12 with another question like the one bellow that is directed to reneges part of the demographic in which they need a place to pray less than 20% of them say they don't have a place to pay in there working place which is once again 20 to 80 law that makes is a minority

Figure Results and discation .12 my workplace has a specific place to pray

Another controversial question is having bounces or resigning a boost on salary for doing work over hours or over time in most other terms again they are divided in to 3 parts 1 part of they say they don't another say they get it on the part that is not clear and that may have to do with the works themselves they may not put in extra work for that they don't know if they get an extra pay or not so they have selected nutmeal in the respond as shown in the chart below figure 4.13

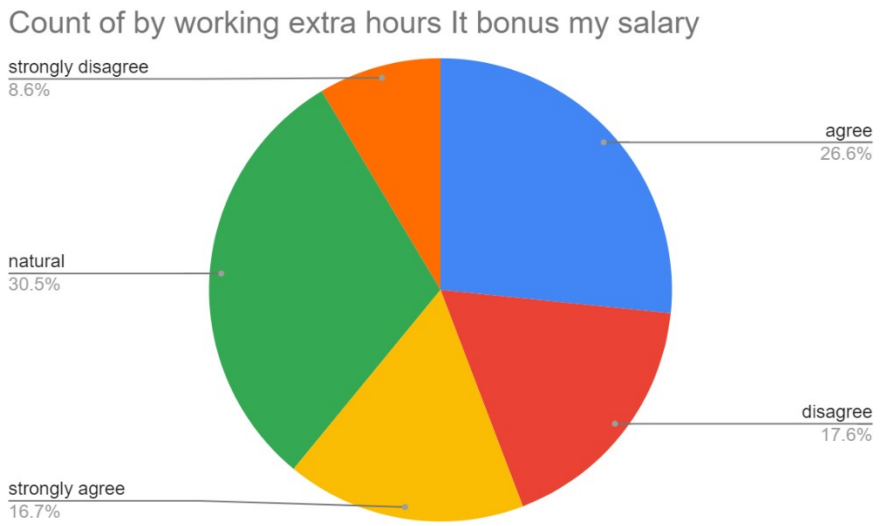

Figure Results and discation .13 by working extra hours It bonus my salary

Figer 4.14 respondents' views on whether their type of work has caused any family problems. The largest portion, at 36.5%, indicated disagreement, suggesting that a significant number of participants do not feel their work has negatively impacted their family life. Following this, 24.1% were neutral on the issue. However, a considerable 22.4% agreed that their work had caused family problems, and 10.8% strongly disagreed, indicating a mixed set of responses overall. A smaller segment, 7.3%, strongly agreed, highlighting that while the majority don't perceive work as causing family problems, a notable minority does.

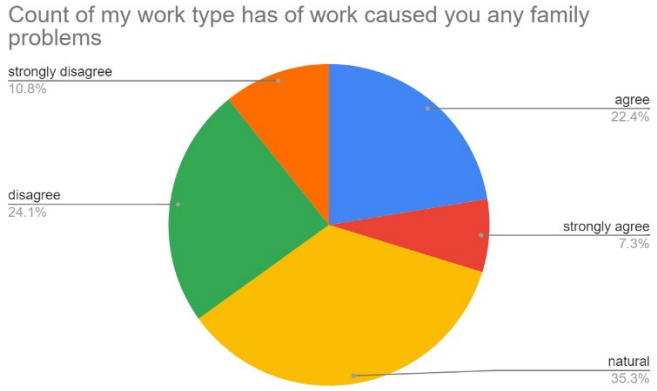

Figure Results and discation .14 my work type has of work caused you any family problems

Figure 4.15 illustrates the perceived importance of having a safety officer present at work. Most respondents, at 36.6%, agreed on the importance, with a further 27.7% remaining neutral. Interestingly, 23.1% strongly agreed, emphasizing a significant acknowledgment of the safety officer's role. On the other hand, 10.1% disagreed and 2.7% strongly disagreed, indicating that

while most participants recognize the importance of a safety officer, a small portion do not see it as crucial.

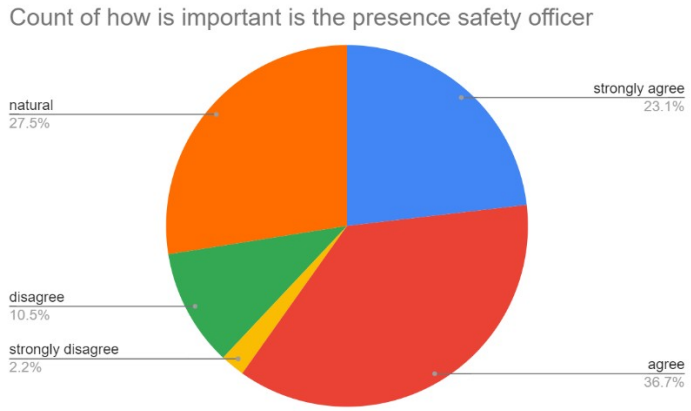

Figure Results and discation .15 how is important is the presence safety officer

Figure16depicts respondents' satisfaction with their work-life balance. The highest percentage, 31.9%, agreed that they are satisfied with their work-life balance. Another significant segment, 31.7%, strongly disagreed, indicating a notable dissatisfaction among some participants. 17.1% disagreed, while 13.5% were neutral on the issue. A smaller portion, 5.8%, strongly agreed, suggesting that while some workers are content with their work-life balance, others face considerable dissatisfaction.

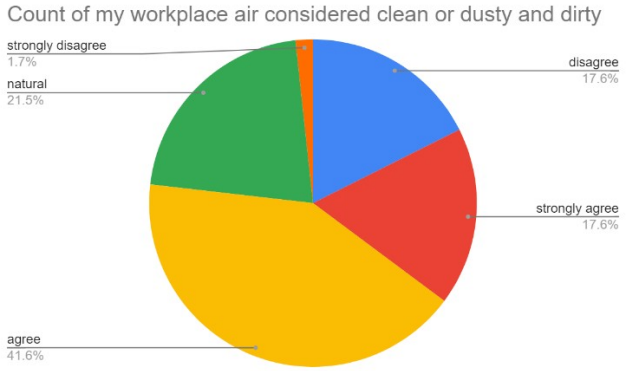

Figure Results and discation .16 I am satisfied with my work life balance?

Figure 4.16 pie chart shows respondents' perceptions of injury occurrences at their workplace. The largest segment, at 37.7%, is neutral, indicating uncertainty or mixed experiences. A significant 24.7% agree that injuries occur, while 13.5% strongly agree, highlighting concerns about workplace safety. In contrast, 16.7% disagree and 7.4% strongly disagree, suggesting that some believe their workplace is relatively safe

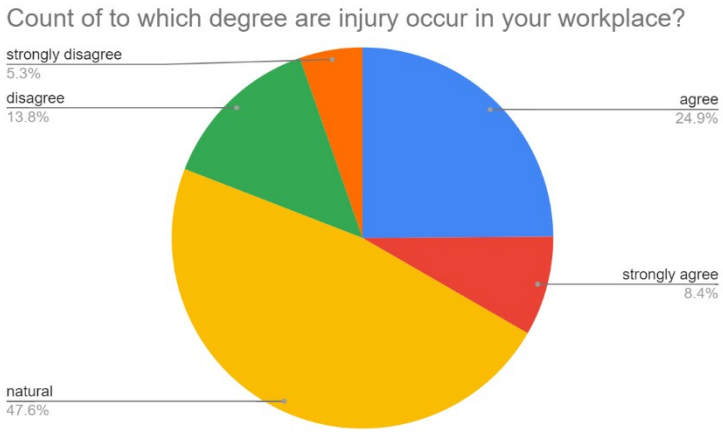

Figure Results and discation .17 to which degree are injury occur in your workplace?

how safe respondents feel their workplace is. The majority, at 36.9%, agree that their workplace is safe. Another 29.2% are neutral, while 19.2% strongly agree, indicating a general sense of safety among many respondents. However, 8.1% disagree and 6.7% strongly disagree, showing that some employees do not feel safe. from figure 4.17

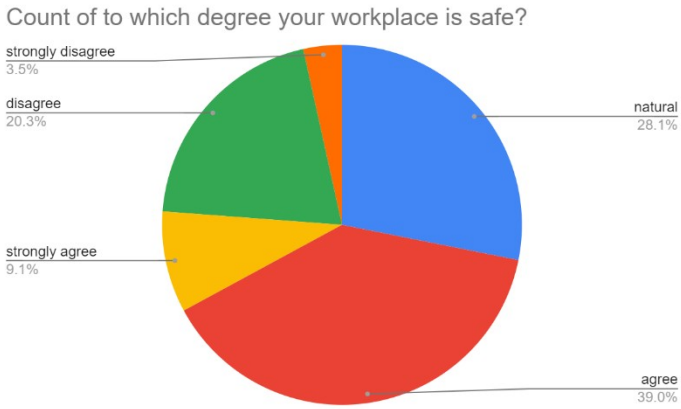

Figure Results and discation .18 to which degree your workplace is safe?

spendings' views on managerial motivation for creating a suitable work environment. The largest portion, at 39.5%, is neutral. A significant 28.3% agree and 10.7% strongly agree, indicating positive managerial influence for some. On the other hand, 15.8% disagree and 5.8% strongly disagree, highlighting areas for improvement in managerial support. As shown in figure 4.18

Figure Results and discation .19 my manager motivates me to create a suitable work environment

Figure 4.19 states that respondents' feelings about having excess time. The largest segment, at 45.0%, is neutral, indicating a balanced view on this issue. A significant 23.4% agree and 8.1% strongly agree, suggesting that some feel they have too much time. Conversely, 23.8% disagree and 1.7% strongly disagree, indicating that others do not feel they have excess time.

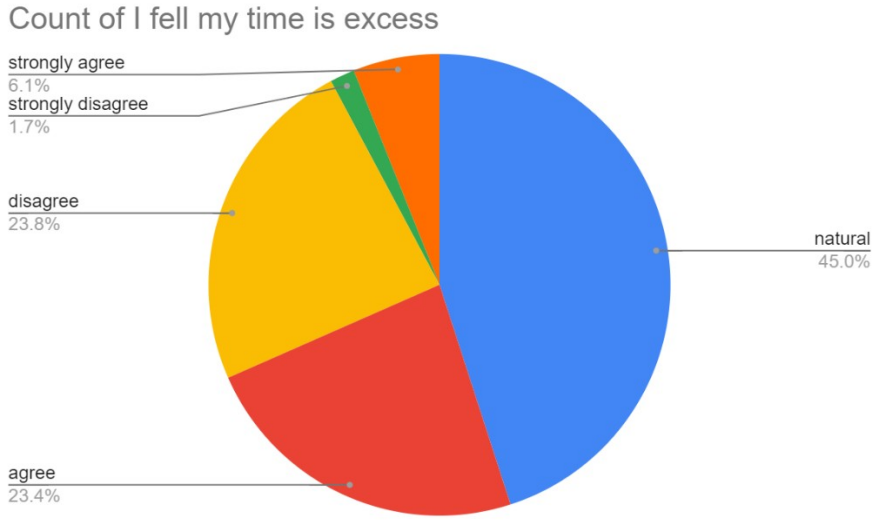

Figure Results and discation .20 I fell my time is excess

how often respondents feel they have less energy. The largest portion, at 45.0%, is neutral. A notable 29.3% agree and 3.0% strongly agree, indicating that a considerable number of respondents frequently feel low on energy. On the other hand, 21.1% disagree and 1.6% strongly disagree, suggesting that some do not often experience low energy levels. From figure 4.20

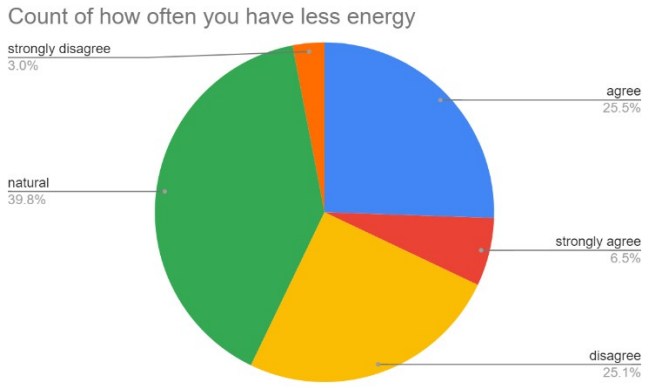

Figure Results and discation .21 how often you have less energy

In figure 4.21 feelings about experiencing strain while working. The largest segment, at 39.2%, agrees that they feel an excess of strain, followed by 18.1% who strongly agree. Conversely, 17.7% disagree and 3.0% strongly disagree, indicating a portion of respondents do not feel significant strain. The remaining 22.0% are neutral on the issue.

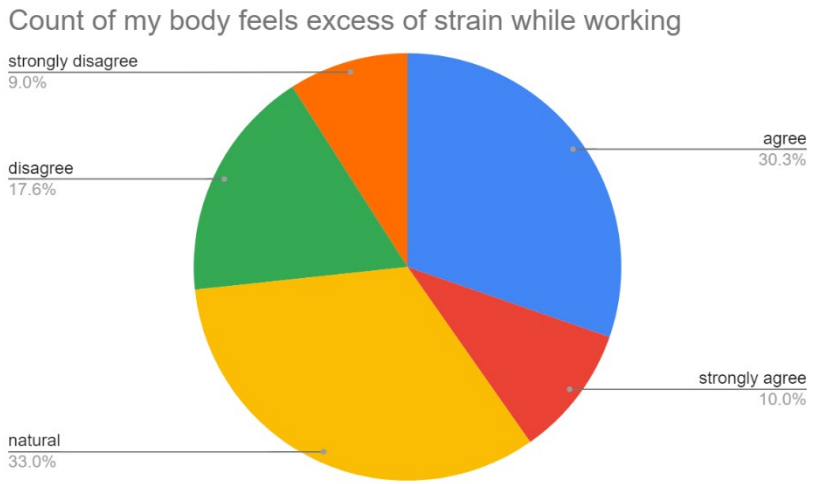

Figure Results and discation .22 my body feels excess of strain while working

He degree of instruction from supervisors or safety officers while working. The largest portion, at 30.9%, agrees that they receive adequate instruction. Meanwhile, 27.5% are neutral, and 21.5% disagree, indicating some lack of instruction. Additionally, 17.7% strongly agree and 2.3% strongly disagree, highlighting mixed levels of perceived guidance as shown in figure 4.22

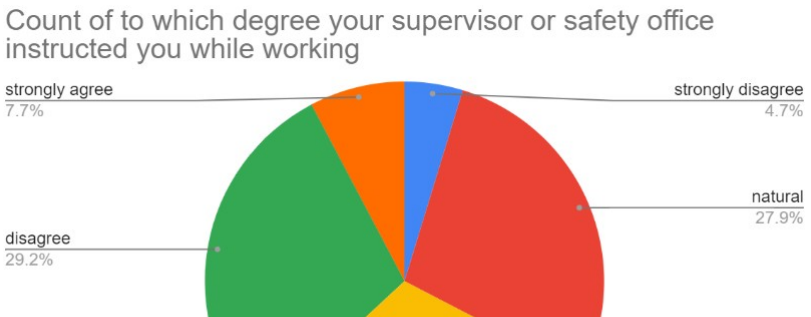

Figure Results and discation .23 to which degree your supervisor or safety office instructed you while working

In figure 4.23 views on the suitability of their workplace environment. The majority, at 36.5%, agree that their environment is suitable, while 23.7% are neutral. Conversely, 22.7% disagree and 3.6% strongly disagree, indicating some dissatisfaction with the work environment. A notable 13.5% strongly agree, showing a range of experiences among respondents.

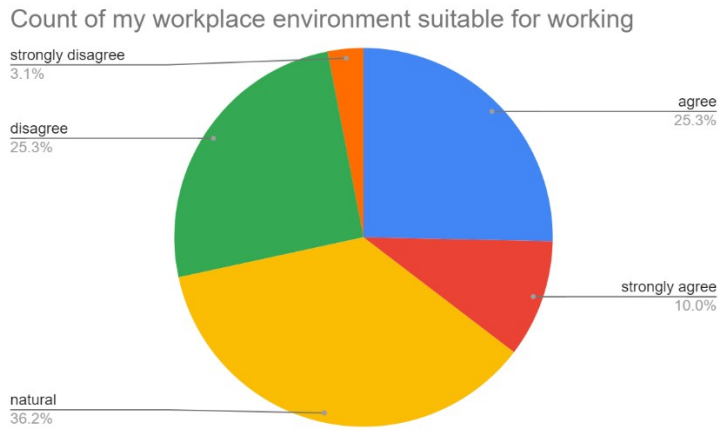

Figure Results and discation .24 my workplace environment suitable for working

forward protection tradition from figure 4.24 A safety tradition that integrates psychosocial and ergonomic considerations fosters a holistic technique to safety. people are much more likely to prioritize safety and engage in proactive behaviors after they see that their proper-being is a top precedence for the company elevated productiveness and efficiency: Ergonomic interventions that lessen bodily pressure can decorate productiveness and efficiency. workers who're comfortable and pain-free are much more likely to carry out their tasks efficiently and with extra precision

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

3.3 Conclusion

To conducting this study we are found out that most of the workers that I work in construction sites tend to often feel muscle joint pains and they are often way much often lifting heavy weights but they lack any sort of Training or any sort of procedure on how to actually lift these weights in a way that they don't feel any sort of pain and in a way that's safe and scientific in order to maintain a high psychological value of these workers while there's about two third of them saying they feel muscle pain often less than quester saying they don't feel that much pain in muscles and joints this indicates that they are have you workplace environment. The majority, at 36.5%, agree that their environment is suitable, while 23.7% are neutral. Conversely, 22.7% disagree and 3.6% strongly disagree, indicating some dissatisfaction with the work environment. A notable working with materials that are not easy to move around with and the mobility of these items from them often bent much in order to lift these weights off the ground and they tend to develop or this can manifest into muscle pain. Then over all the are satisfied with their work and their life and their control over there comment and they are mostly comfortable on the only thing that we can indicate from this study is those three or four things that I have mentioned before this section most of the data tends to be towards the positive side and towards the national side so we can say these workers are mostly psychologically and economically satisfied with their work

3.4 recommendation

1. Implement support programs including stress management workshops and problem-solving training.
2. Foster a positive organizational culture through open communication, regular feedback, and employee recognition.
3. Enhance resource allocation by regularly assessing needs and investing in ergonomic tools and technology.
4. Promote work-life balance with flexible working hours, remote work options, and encouraging regular breaks.

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